

Phase 2 of Traffic light controller project

NOTE: This is just a planning flow that might be changed during development, Phase 2 is currently under development.

Existing System

file name and their work

1. timer.v - Generate time, 1 clk cycle is equal to 1 second.
2. state_register.v - Store the current state of the system.
3. Next_state.v - Based on current state and inputs, decide the next state and set the priority.
4. output_logic.v - Based on the current state, decide which output pin should go high and low.
5. traffic_light_controller.v - use all the modules to build the system (top level module)

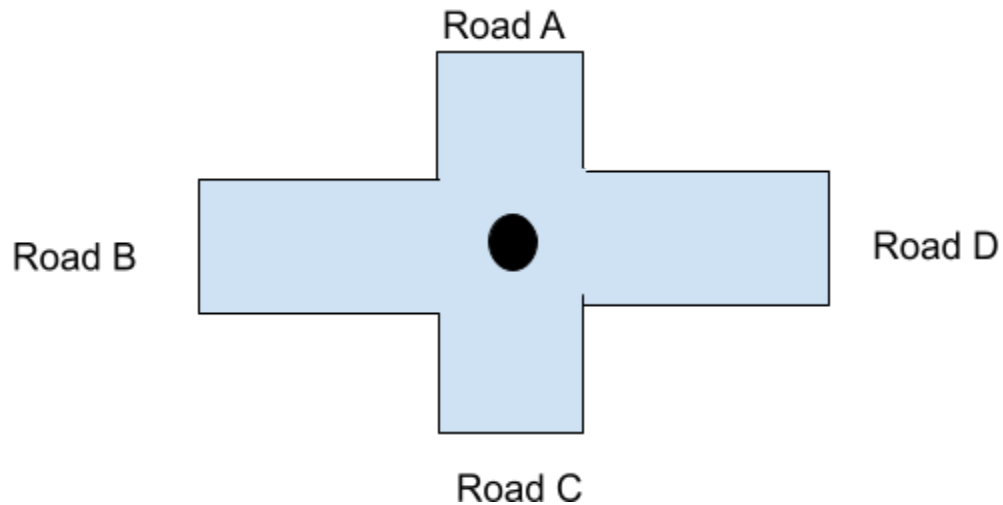
State table

State	Name
S0	NS_Green
S1	NS_Yellow
S2	All_red_NS
S3	walk_NS
S4	Emergency_NS
S5	EW_Green
S6	EW_yellow
S7	All_red_EW
S8	Walk_EW
S9	Emergency_EW

Phase 2 -

Next System Requirements

1. Four-Way traffic management
2. Expanded FSM architecture
3. Additional pedestrian crossings
4. Emergency handling for four direction
5. More advanced traffic sequencing.



Input and output of the system

- **Inputs**
 - Clk
 - Reset
 - emergency_A
 - ped_req_A
 - emergency_B
 - ped_req_B
 - emergency_C
 - ped_req_C
 - emergency_D
 - ped_req_D
- **Outputs**
 - Red_A
 - Green_A
 - Yellow_A
 - wlak_A
 - dont_walk_A
 - Red_B
 - Green_B
 - Yellow_B
 - wlak_B
 - dont_walk_B

- Red_C
- Green_C
- Yellow_C
- wlak_C
- dont_walk_C
- Red_D
- Green_D
- Yellow_D
- wlak_D
- dont_walk_D

(Total 10 input signal and 20 output signal)

State table

State	Name
S0	A_Green
S1	A_Yellow
S2	All_red_A
S3	walk_A
S4	Emergency_A
S5	B_Green
S6	B_yellow
S7	All_red_B
S8	Walk_B
S9	Emergency_B
S10	C_Green
S11	C_Yellow
S12	All_red_C
S13	walk_C
S14	Emergency_C

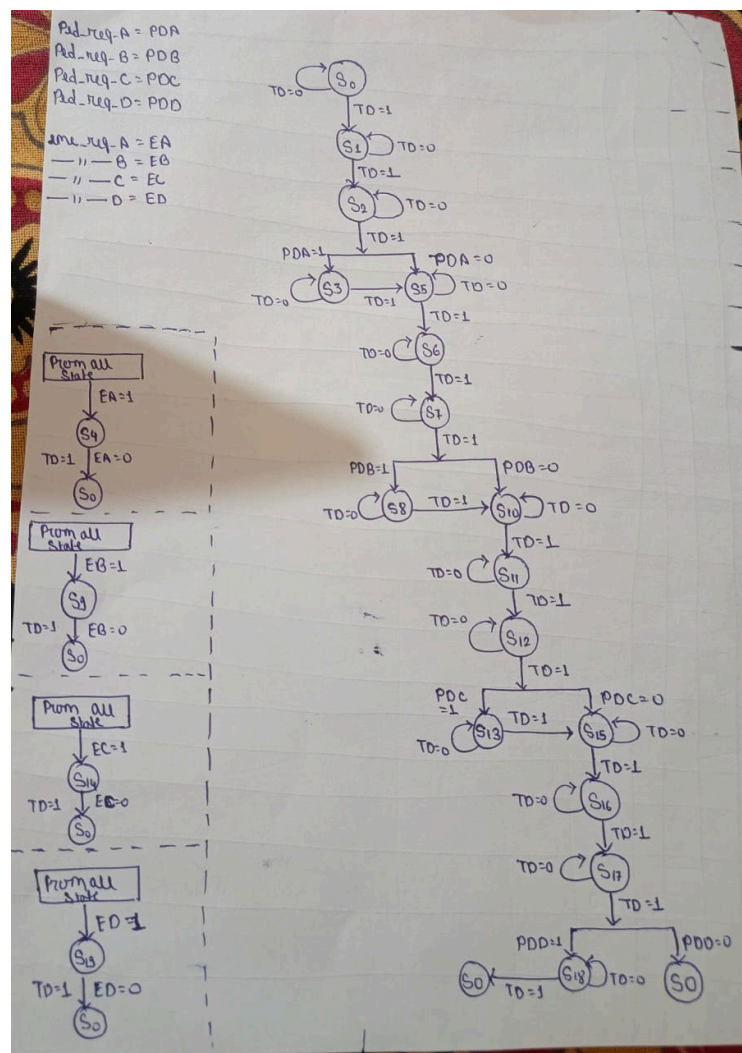
State	Name
S15	D_Green
S16	D_Yellow
S17	All_red_D
S18	walk_D
S19	Emergency_D

Total 20 states form S0 to S19

Important points

1. Normal system flow $S0 > S1 > S2 > S5 > S6 > S7 > S10 > S11 > S12 > S15 > S16 > S17 > S0$. (With no ped req and emergency request).
2. Ped request should check on state S2, S7, S12, S17.
3. Emergency requests should check on every state with priority $A > B > C > D$.

State diagram



Output Truth table

State	A_red	A_green	A_yellow	A_walk	A_dont_walk	B_red	B_green	B_yellow	B_walk	B_dont_walk	C_red	C_green	C_yellow	C_walk	C_dont_walk	D_red	D_green	D_yellow	D_walk	D_dont_walk
S0	0	1	0	0	1	1	0	0	0	1	1	0	0	0	1	1	0	0	0	1
S1	0	0	1	0	1	1	0	0	0	1	1	0	0	0	1	1	0	0	0	1
S2	1	0	0	0	1	1	0	0	0	1	1	0	0	0	1	1	0	0	0	1
S3	1	0	0	1	0	1	0	0	0	1	1	0	0	0	1	1	0	0	0	1
S4	0	1	0	0	1	1	0	0	0	1	1	0	0	0	1	1	0	0	0	1
S5	1	0	0	0	1	0	1	0	0	1	1	0	0	0	1	1	0	0	0	1
S6	1	0	0	0	1	0	0	1	0	1	1	0	0	0	1	1	0	0	0	1
S7	1	0	0	0	1	1	0	0	0	1	1	0	0	0	1	1	0	0	0	1
S8	1	0	0	0	1	1	0	0	1	0	1	0	0	0	1	1	0	0	0	1
S9	1	0	0	0	1	0	1	0	0	1	1	0	0	0	1	1	0	0	0	1
S10	1	0	0	0	1	1	0	0	0	1	0	1	0	0	1	1	0	0	0	1
S11	1	0	0	0	1	1	0	0	0	1	0	0	1	0	1	1	0	0	0	1
S12	1	0	0	0	1	1	0	0	0	1	1	0	0	0	1	1	0	0	0	1
S13	1	0	0	0	1	1	0	0	0	1	1	0	0	1	0	1	0	0	0	1
S14	1	0	0	0	1	1	0	0	0	1	0	1	0	0	1	1	0	0	0	1
S15	1	0	0	0	1	1	0	0	0	1	1	0	0	0	1	0	1	0	0	1
S16	1	0	0	0	1	1	0	0	0	1	1	0	0	0	1	0	0	1	0	1
S17	1	0	0	0	1	1	0	0	0	1	1	0	0	0	1	1	0	0	0	1
S18	1	0	0	0	1	1	0	0	0	1	1	0	0	0	1	1	0	0	1	0
S19	1	0	0	0	1	1	0	0	0	1	1	0	0	0	1	0	1	0	0	1

Files need to modified in phase 2

1. State_register_logic.v
2. Next_state_logic.v
3. Output_logic.v
4. traffic_light_controller.v

(also their respective TB for module testing)